Ramrao Adik Institute of Technology, Nerul, Navi Mumbai

Question Bank – 4 (Module-4)

Academic year: 2025-2026

Subject: Embedded System Class: S.E. Sem : III

1. List and explain all addressing modes in 8051 microcontroller.
2. Explain how I/O ports are initialized in 8051 with suitable instructions and examples.
3. Explain MUL and DIV instructions in 8051 with suitable examples.
4. Explain ANL, ORL and XRL instructions in 8051 with suitable examples.
5. Explain RL, RLC, RR, RRC instructions in 8051 with suitable examples.
6. Explain TMOD and TCON SFR's of 8051 microcontroller in detail.
7. Explain how 8051 can be used in Counter Mode.
8. Explain how 8051 serial communication works with suitable SFR's.
9. Explain what RS232 is and hence explain DTE and DCE too.
10. Explain the 8051 interrupts with Interrupt Vector table (IVT) in detail.
11. Write a assembly language program to find the largest number from given five numbers.
12. Write a assembly language program to unpack a packed BCD Number.
13. Write a assembly language program to generate a delay of 1ms.
14. Write a program for the 8051 to transfer letter "A" continuously on the serial port.
15. Write a program to detect the position of switch connected to P2.7 pin and accordingly ON and OFF the LED connected at pin P1.0. LED should be ON when the switch is in ON position and LED should be OFF when the switch is in OFF position.